

An Empirical Framework for the Health-Cost Incidence, Persistence, and Distributional Burden of Climate Shocks

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1 Setting and identification

The analysis uses two panels that share a common county-level climate-shock series but measure different outcome domains. *Panel A* is a county $\times$ year panel (demand side): $\sim 41,400$ county-years, 3,155 counties, 2011–2023, with outcomes for individual-market benchmark premiums, medical-debt share, real personal income per capita, and civilian employment. *Panel B* is a hospital $\times$ year panel (supply side): $\sim 59,900$ hospital-years, 5,119 hospitals (CMS Certification Numbers), with operating and net margins and uncompensated care. Each hospital is geo-coded to a single county (modal county per ZIP, facility location) and inherits that county’s shocks, so $S_{it} \equiv S_{c(i),t}$. Both panels carry county (or hospital) and year fixed effects and cluster standard errors at the state level (~ 50 clusters).

All papers use four binary shocks defined at the county-year level: `Is_Extreme_Drought` (Palmer PDSI ≤ -4); `High_CDD / High_HDD` (cooling/heating degree-days in the top quartile of the county’s own historical distribution); and `High_AQI_Max` (annual maximum AQI > 100). Generically $S_{ct} \in \{0, 1\}$. These are *recurring* shocks, not one-time absorbing treatments: a county may enter, leave, and re-enter drought. This rules out canonical staggered-adoption DiD as the *primary* design and motivates the persistence estimators of Section 3; Section 2 adds a first-onset cohort DiD as a contamination-free cross-check. Defining shocks against each county’s own 1990–2000 baseline distribution makes “shock” an unusual draw relative to local historical norms rather than an absolute threshold.

2 Paper 1 — Incidence: do climate shocks raise health costs?

2.1 Distributed-lag two-way fixed effects

Paper 1 asks whether contemporaneous and lagged exposure shifts the *level* of the outcome. For outcome Y and shock S ,

$$Y_{ct} = \alpha_c + \gamma_t + \sum_{k=0}^2 \beta_k S_{c,t-k} + \mathbf{X}'_{ct} \delta + \varepsilon_{ct}, \quad (1)$$

with county FE α_c , year FE γ_t , and time-varying controls $\mathbf{X}_{ct}$ (log real income per capita, unemployment, Medicaid-expansion status); the hospital version replaces c with hospital i . Equation (1) is estimated *once per (outcome, shock) pair* on the full panel; the three lag terms enter the same regression. The subscripts run $t, t-1, t-2$ because a shock can only affect current and *subsequent* outcomes: a single drought in year t appears as the β_0 regressor for the year- t outcome, the β_1 regressor for the year- $(t+1)$ outcome, and the β_2 regressor for year $t+2$. The identifying variation is counties that *switch* shock status; always- and never-shocked counties are absorbed by α_c . The cumulative (impulse-response) effect of one shock-year traced across all three horizons is

$$\hat{\beta}_{\text{cum}} = \hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2, \quad \widehat{\text{Var}}(\hat{\beta}_{\text{cum}}) = \sum_{j=0}^2 \sum_{k=0}^2 \widehat{\text{Cov}}(\hat{\beta}_j, \hat{\beta}_k), \quad (2)$$

the off-diagonal covariances taken from the state-clustered vcov via a Wald linear-combination test. On the hospital panel, heat raises uncompensated care (High_CDD: +1.22 pp of net patient revenue, $p = 0.063$; cold: +\$1.55M per hospital, $p = 0.022$). The negative drought coefficient on uncompensated care (-1.68 pp, $p = 0.032$) is a measurement artifact—billing and credit-bureau measures undercount distress in drought-stricken, lower-utilization areas—and is flagged throughout rather than read causally.

2.2 A sharper control group: the 2012 drought natural experiment

Because the lag model’s control group for any drought year silently includes counties droughted in *other* years, we add a first-onset cohort DiD. Let $\text{Treated}_c = 1$ if county c ’s first-ever extreme-drought year in the panel is 2012, and $\text{Post}_t = \mathbf{1}[t \geq 2012]$:

$$Y_{ct} = \alpha_c + \gamma_t + \tau (\text{Treated}_c \times \text{Post}_t) + \varepsilon_{ct}. \quad (3)$$

The FEs absorb both main effects, so $\hat{\tau}$ is the ATT: 139 first-onset-2012 counties against 2,534 *never*-droughted controls (counties first droughted in other years are excluded, which is what cleans the control group). With a single pre-period (2011) the regression returns the 2×2 difference exactly.

The sharp 2×2 assumes *unconditional* parallel trends, which is strong here: treated counties skew rural-agricultural (Georgia, the Mountain West, the Plains) while controls skew urban and coastal. We therefore also report the Sant’Anna–Zhao (2020) doubly-robust ATT, consistent if *either* the propensity or the outcome-regression model is correct, conditioning only on *baseline* (2011) covariates—log population, baseline median household income, Census division—and deliberately not on contemporaneous income, employment, or premiums, which are post-treatment mediators (bad controls):

$$\hat{\tau}_{\text{dr}} = \mathbb{E} \left[\left(\frac{D}{\mathbb{E}[D]} - \frac{\hat{\pi}(X)(1-D)/(1-\hat{\pi}(X))}{\mathbb{E}[\hat{\pi}(X)(1-D)/(1-\hat{\pi}(X))]} \right) (\Delta Y - \hat{m}_0(X)) \right] \quad (4)$$

where ΔY is the pre-to-post outcome change (post collapsed to the 2012–2023 mean), $\hat{\pi}(X)$ the propensity score, and $\hat{m}_0(X)$ the control outcome-change regression. Results are in Table 1.

Table 1: Incidence: 2012-drought natural experiment (county panel, \$2023)

Outcome	Uncond. 2×2 $\hat{\tau}$	Doubly-robust $\hat{\tau}_{\text{dr}}$	DR 95% CI
Personal income per capita	−1,311	− 1,451	[−2,461, −441]
Civilian employed (persons)	−2,053	−871	[−1,719, −23]
Median household income	−992	−1,186	[−2,140, −232]
Medical-debt share	−0.006 (n.s.)	−0.011	[−0.019, −0.002]

139 treated (first-onset 2012) vs. 2,534 never-exposed; state-clustered. 2×2 : income $z = -2.28$ ($p = 0.027$), employment $z = -4.29$ ($p = 0.0001$). Income is robust to conditioning (if anything stronger); employment attenuates $\sim 58\%$ and is the composition-sensitive secondary finding. Medical debt is measurement-fragile. Re-estimated as a full Callaway–Sant’Anna model across *all* drought cohorts the average is null (PCPI +\$350, s.e. 585), so the negative real-economy effect is specific to the large, clean 2012 event and is an intent-to-treat “effect of first onset.”

3 Paper 2 — Persistence: scarring, reversibility, dose

3.1 Onset/persist/exit local projections

Converting the binary series into a transition state from $(S_{ct}, S_{c,t-1})$ gives *onset* $D^{on} = (1, 0)$, *persist* $D^{per} = (1, 1)$, and *exit* $D^{ex} = (0, 1)$, with $(0, 0)$ omitted as the reference. Following Jordà (2005), a separate regression is run per horizon h :

$$Y_{i,t+h} = \alpha_i + \gamma_t + \beta_h^{on} D_{it}^{on} + \beta_h^{per} D_{it}^{per} + \beta_h^{ex} D_{it}^{ex} + \varepsilon_{i,t+h}, \quad h = 0, 1, 2, \quad (5)$$

where the LHS is the outcome h years after the transition year t and the dummies are always indexed by t . Each coefficient is measured against the never-shocked $(0,0)$ reference: β_h^{on} is the entry effect, β_h^{per} the interior of a sustained spell, and β_h^{ex} the first post-shock year. If a shock were fully reversible the entry cost would be exactly undone on exit, so the null and the asymmetry statistic are

$$H_0 : \beta_h^{on} + \beta_h^{ex} = 0, \quad \hat{L}_h = \hat{\beta}_h^{on} + \hat{\beta}_h^{ex}, \quad z_h = \frac{\hat{L}_h}{\sqrt{V_{11} + V_{33} + 2V_{13}}} \xrightarrow{d} \mathcal{N}(0,1), \quad (6)$$

the covariance V_{13} taken directly from the single fitted model's vcov. A nonzero $\hat{L}_h$ means exit does not return the unit to its pre-shock trajectory—scarring.

3.2 Cumulative dose

A second lens asks whether the *stock* of past exposure matters. Define a never-resetting counter $C_{it} = \sum_{s \leq t} S_{is}$ and estimate

$$Y_{it} = \alpha_i + \gamma_t + \beta_1 C_{it} + \beta_2 C_{it}^2 + \varepsilon_{it}, \quad \left. \frac{\partial Y}{\partial C} \right|_{C=x} = \beta_1 + 2\beta_2 x, \quad (7)$$

with a binned variant $\sum_b \delta_b \mathbf{1}[C_{it} \in b]$ over $b \in \{1-3, 4-6, 7-9, 10+\}$ (reference $C = 0$) and a high-vs-low contrast $\hat{\Delta} = \hat{\delta}_{10+} - \hat{\delta}_{1-3}$. Table 2 reports drought $\rightarrow$ hospital operating margin.

Table 2: Persistence: extreme drought $\rightarrow$ hospital operating margin ($N = 57,302$)

Quantity	Estimate	Test
$\hat{\beta}_0^{on}$ (onset)	+0.0159	margin +1.59 pp vs. $(0,0)$
$\hat{\beta}_0^{per}$ (persist)	-0.0017	—
$\hat{\beta}_0^{ex}$ (exit)	+0.0109	still +1.09 pp after clearing
Asymmetry $\hat{L}_0 = \hat{\beta}_0^{on} + \hat{\beta}_0^{ex}$	+0.0268	SE 0.0098, $z = 2.73$, $p = 0.006$
Dose marginal at $x = 5$ (Eq. 7)	+0.0023	$p = 0.010$
Dose marginal at $x = 10$	+0.0018	$p = 0.14$
Binned $\hat{\Delta} = \hat{\delta}_{10+} - \hat{\delta}_{1-3}$	+0.018	$p = 0.033$

Onset and exit are *both* positive: the margin does not unwind on exit (H_0 rejected at 1%)—asymmetric persistence. The positive dose-response is read as survivorship/adaptation (financially weaker hospitals exit the unbalanced panel first), not as climate exposure improving finances; the persistence claim is anchored to the drought hysteresis result, which needs no long accumulation window.

4 Paper 3 — Inequality and provider heterogeneity

Paper 3 asks whether the effects of Papers 1–2 concentrate in already-disadvantaged groups. For a moderator M_{it} ,

$$Y_{it} = \alpha_i + \gamma_t + \beta S_{ct} + \theta(S_{ct} \times M_{it}) + \phi M_{it} + \mathbf{X}_{it}' \delta + \varepsilon_{it}, \quad (8)$$

so β is the shock effect at $M = 0$, $\beta + \theta$ at $M = 1$ (SE from $\sqrt{V_{\beta\beta} + V_{\theta\theta} + 2V_{\beta\theta}}$), and θ the differential. The demand side uses county social vulnerability (CDC/ATSDR SVI, top-quartile dummy); the supply side uses hospital safety-net status, ownership, state Medicaid-expansion, and county hospital-market concentration (HHI). For strain-increasing outcomes (uncompensated-care %) $\theta > 0$ indicates concentration in high- M units; for operating margin a negative θ does. Table 3 gives selected interactions.

5 Inference

All standard errors cluster at the state level throughout; for HIX-premium outcomes the county panel additionally reports rating-area-clustered SEs, since counties sharing a rating area have identical premiums by construction. Every post-estimation quantity—cumulative lag effects (2), asymmetry statistics (6),

Table 3: Heterogeneity: selected shock $\times$ moderator interactions

Shock $\times$ moderator	Outcome	Direction of $\hat{\theta}$	p
High_CDD $\times$ SafetyNet	Uncomp. care %NPR	higher for safety-net	0.021
Drought $\times$ SVI (county)	Income per capita	larger loss in high-SVI	< 0.05
High_HDD $\times$ SafetyNet	Uncomp. care %NPR	higher for non-safety-net	0.011
Drought $\times$ Medicaid expansion	Uncomp. care %NPR	higher in expansion states	< 0.001

Headline: heat-driven uncompensated care concentrates in safety-net hospitals—the supply-side analogue of high-SVI county amplification on the demand side. Interactions are shock- and outcome-specific and not all point the “vulnerability amplifies” way; all are reported transparently.

dose-level marginal effects (7), and moderator marginal effects (8)—is a linear combination $\hat{L} = \mathbf{w}'\hat{\beta}$ with $\widehat{\text{Var}}(\hat{L}) = \mathbf{w}'\hat{V}\mathbf{w}$ evaluated on the state-clustered vcov $\hat{V}$; tests use the normal approximation and are two-tailed at $\alpha = 0.05$. All models are estimated with `fixest::feols` (Bergé 2018), which performs the two-way within transform and exports the clustered vcov for the `lincom` steps.

Appendix: estimating-equation reference

Paper	Estimating equation	Key quantity
1 Incidence	$Y_{it} = \alpha_i + \gamma_t + \sum_{k=0}^2 \beta_k S_{i,t-k} + \varepsilon$	$\hat{\beta}_{\text{cum}} = \sum_k \hat{\beta}_k$
1 Incidence (DiD)	$Y_{ct} = \alpha_c + \gamma_t + \tau(\text{Tr}_c \times \text{Post}_t) + \varepsilon$	ATT $\hat{\tau} = \Delta^T - \Delta^C$
2 Persistence (onset/exit)	$Y_{i,t+h} = \alpha_i + \gamma_t + \beta^{on} D^{on} + \beta^{per} D^{per} + \beta^{ex} D^{ex} + \varepsilon$	$\hat{L}_h = \hat{\beta}_h^{on} + \hat{\beta}_h^{ex}$
2 Persistence (dose)	$Y_{it} = \alpha_i + \gamma_t + \beta_1 C_{it} + \beta_2 C_{it}^2 + \varepsilon$	$\hat{\beta}_1 + 2\hat{\beta}_2 x$
3 Heterogeneity	$Y_{it} = \alpha_i + \gamma_t + \beta S_{it} + \theta(S_{it} \times M_{it}) + \phi M_{it} + \varepsilon$	differential $\hat{\theta}$

Estimation in R (`fixest`). Shock construction: `process_county_climate.R`, `process_hospital_panel.R`. Persistence helpers: `transition_symmetry.R`, `cumulative_dose.R`. Natural-experiment DiD: `run_did_analysis.R`; doubly-robust check: `did_robustness/02_doubly_robust_did.R`.